Classical Machine Learning Baseline for Skin Lesion Classification

MILK10k dataset · Binary benign/malignant classification

Author: Clara Lacroix

Project repository: https://github.com/claralacroix/milk10k-skin-lesion-classification

Project objective

The objective was to develop and evaluate a classical machine-learning baseline for skin-lesion classification. Hand-crafted image features were extracted and used to compare three standard classifiers and two feature sets.

Main result

The best tested configuration was a Random Forest using combined colour and texture features. It achieved 80.4% accuracy and a macro F1-score of 0.742 on the held-out test split. This is an educational baseline and is not suitable for clinical decision-making.

1. Dataset and preprocessing

Labels were read from lesion_table.csv. The target was diagnosis_1, with the two classes **Benign** and **Malignant**. Indeterminate cases were excluded. Each lesion was represented by an available dermoscopic image whose filename matched derm_id. After filtering for usable images, 5,117 lesions remained: 3,634 malignant and 1,483 benign.

Images were converted to RGB and resized to 128 × 128 pixels. A fixed 80/20 hold-out split was created using GroupShuffleSplit with random_state=42 and lesion_id as the grouping variable. All configurations used the same split so their scores could be compared fairly.

2. Feature sets

Colour only: 16-bin normalized histograms for each red, green, and blue channel (48 features total).

Colour + texture: the same colour histograms, plus grayscale mean, standard deviation, 10th and 90th percentiles, and gray-level co-occurrence matrix (GLCM) properties: contrast, homogeneity, energy, and correlation across two angles.

3. Models and evaluation

Three classifiers were evaluated: Logistic Regression, Support Vector Machine with an RBF kernel, and Random Forest. Class weights were balanced for all models. Logistic Regression and SVM used median imputation and standardization in a pipeline; Random Forest used median imputation. The primary selection metric was macro F1, supported by accuracy and balanced accuracy.

4. Results

Feature set	Model	Accuracy	Balanced acc.	Macro F1
Colour + texture	Random Forest	0.8037	0.7328	0.7419
Colour only	Random Forest	0.7969	0.7248	0.7333
Colour + texture	SVM (RBF)	0.7754	0.7347	0.7262
Colour only	SVM (RBF)	0.7559	0.7100	0.7024
Colour + texture	Logistic Regression	0.7334	0.6957	0.6828
Colour only	Logistic Regression	0.7158	0.6960	0.6734

The combined feature set improved macro F1 for all three models. Random Forest with colour and texture achieved the highest macro F1 (0.7419), so it was selected as the best configuration. SVM with colour and texture had the highest balanced accuracy (0.7347), narrowly above Random Forest.

5. Class-level performance and limitations

For the selected Random Forest configuration, benign precision was 0.66, recall 0.58, and F1 0.62. Malignant precision was 0.85, recall 0.89, and F1 0.87. The classifier performed substantially better on malignant than benign cases in this test split.

Limitations: this is a single hold-out evaluation, and results may vary with a different split. The dataset is imbalanced, and benign recall is limited. The experiment uses hand-crafted features rather than learned image representations, does not establish clinical validity, and should not be used to diagnose patients. Further work could use repeated grouped cross-validation, threshold analysis, calibration, and external validation.

6. Code and reproducibility

The implementation, model comparison table, and confusion-matrix figure are available in the public GitHub repository. The dataset itself is not included in the repository.

Repository: https://github.com/claralacroix/milk10k-skin-lesion-classification

Python experiment script: src/classical_ml_baseline.py

Model comparison CSV: model_comparison.csv

Confusion matrix: confusion_matrix.png

Conclusion

Among the six tested configurations, Random Forest with combined colour and texture features was the best by macro F1. The results support using both colour and texture descriptors for this baseline, while also highlighting the need to improve performance on benign lesions and to evaluate the method more extensively before drawing broader conclusions.